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IARCO 2026 RESEARCH PROPOSAL

The Price of Fairness: How AI-Driven Personalized Pricing Affects Market Equality

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Key Words: Personalized, Pricing, Consumer Surplus, Producer Surplus, Market

Introduction

Imagine attempting to book an urgent flight on your mobile device, only to find that the price was far beyond your budget. Right next door, your neighbor could be calmly booking the same flight, but with a much lower price. This may become the reality of the future economy if personalized pricing became widespread in a hierarchy of sectors from grocery stores to healthcare plans. This essay argues that a future economy that fully adopts personalized pricing ultimately weakens market competitiveness, reduces consumer surplus, and erodes the ideologies of fairness.

Research Problem

Personalized pricing has seen a massive expansion across industries caused by advances in Artificial Intelligence. This practice raises ethical debates about fairness. It also creates a disruption in market competitiveness between businesses and consumers. While firms assert this pricing model provides the benefits of efficiency and fairness, opposers argue it may weaken market competitiveness, reduce consumer surplus, and lead to unequal economic outcomes. This proposal examines whether such pricing mechanisms undermine competitive markets, erode consumer surplus, and generate systematic economic inequality.

Research Question: In what ways does AI-driven personalized pricing cause instability in the economic market, ideologies of fairness, and systematic economic equality?

Supporting Questions:

1. How does personalized pricing change consumer and producer surplus in elastic vs inelastic markets?
2. To what extent do AI algorithms create economic discrimination that challenges equality?
3. What are the long term effects of personalized pricing for market competition and monopolistic formations?

Methodology

This research proposal primarily relies on secondary data analysis through reputable sources to evaluate the economic effects of personalized pricing. Peer-reviewed Academic articles, industry analyzes, and documented case studies were synthesized to answer this research question. The impacts of consumer surplus, producer surplus, and market power was a large part of this research topic and drives the analytical methodology of whether personalized pricing helps or harms market efficiency and consumer welfare. This analysis synthesizes existing evidence to determine whether personalized pricing generates improvements or redistributes fairness in ways that may justify regulatory intervention.

Existing Literature

Personalized pricing is a strategy companies use to tailor prices to individuals based on personal data and the consumers necessities. Businesses are charging different prices on elastic and inelastic goods to different segments of customers [1]. The approach of changing prices to meet consumer status was traditionally used generations ago to profit businesses [5], but is now widely adopted due to the innovation of Artificial Intelligence [1]. The surge of AI technology allowed companies to use a predictive algorithm to personalize their prices faster and more effectively to the consumer. Each product in the online market could be charged an entirely

distinct variable amount depending on the consumer's demographic profile, willingness to pay, browsing history, and AI's predictive algorithm. AI is fundamentally used in this process to alter its predictive algorithm towards customers based on the data it collects, and decides the price to configure to each individual [6].

Dynamic pricing algorithms increase prices for those who spend more and decrease prices for those who spend less; this was allegedly created due to the illusion of fairness. Others had different perspectives about the concept and various concerns on ideologies like ethicality, fairness, and rights [8]. Additionally, urgency to buy the product increases price while a longer time period decreases the price. This effect is due to a higher willingness to pay if less time is given. When purchasing something urgent, consumers are often more likely to accept higher fares than if they were given less time [11].

Documented Case Studies

One example of a company that has integrated personalized pricing minorly into its products was Coca-Cola. In the summers of 2012 and 2013 [2], Coca-Cola created a vending machine to distribute coke bottles and price them differently depending on the temperature and consumer demand. After consumers found out about this strategy, it caused public resistance towards the company and the CEO was ousted. It is not only Coca-Cola that has attempted to integrate this procedure [1]. Another major player in the game of personalized pricing is Delta Airlines. Delta majorly contributed to the personalization of their pricing strategy by altering their algorithm so it would take into account the consumer's browsing history, device, time, demographics, and consumer loyalty. Delta controlled extensive first-party data, surpassing competitors. This procedure caused privacy and ethical concerns, due to the lack of transparency on which data the AI uses [10]. In theory; If consumers had lots of data, the AI algorithm would determine whether they would pay more or less and how willing they would be to pay this price. If the data-rich consumer had previously purchased many expensive tickets, it would cause them a disadvantage because AI would charge them a higher price this time as well. If the same consumer previously purchased cheaper tickets, the algorithm would offer a cheap fare in this situation, benefitting the consumer. On the other hand, for consumers that did not have much data to provide, the AI algorithm would use data from outside sources such as market competitiveness and the pricing of other companies to decide fares. (Fig. 1)

AI Algorithm	High amount of data	Low amount of data
Cheaper buying	Lower prices	Affected by outside market
Expensive buying	Higher prices	Affected by outside market

Economic Analysis

Personalized pricing does not only affect consumers and can have a large impact on businesses and the overall economy as well. Because such airlines adapted their prices within minutes and to different consumers, business models of travel wholesalers were disrupted. Travel wholesalers such as Expedia and Google flights are dependent on fixed fares and Global Distribution System

(GDS) data [10]. When AI changes prices for flights in real time, these channels often are not able to adapt as quickly to provide consumers with the latest pricing changes [1].

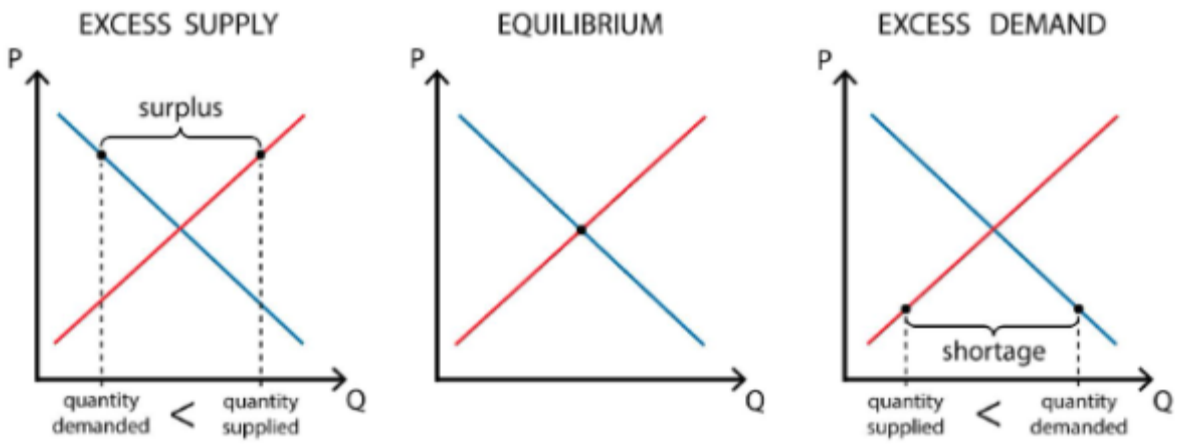
Personalized pricing changes drastically affect the economy through fluctuating the price elasticity of demand. The price elasticity of demand—how elastic a product is based on the demand—can be represented using this formula (*Price Elasticity of Demand = Percentage Change in Quantity Demanded / Percentage Change in Price*). Price elasticity of demand is heavily influenced by current market situations and the substitutability of products [7].

As a result of personalized pricing, rates for elastic products could decrease, causing more people to buy the product; and vice versa. Either way, this could become the foundation for a substantial market shift. Conversely, if prices for an inelastic product fluctuate, the demand would remain the same. Despite this, it is still possible for one company to gain power over consumers, causing the market to shift from competitive to a monopoly [13]. To connect to product elasticity, consumer and producer surplus disruptions also result in significant market changes. Consumer surplus is the actual price of the product subtracted from the consumer's willingness to pay. Similarly, the producer surplus is the market price minus the marginal cost of creating and selling the product.

In today's economy, if a producer could charge the maximum each consumer would accept—exactly their willingness to pay—it would benefit the producer and maximize their margin [1]. This, in turn, would eliminate consumer surplus entirely. In contrast, when consumers purchase goods for less than their maximum willingness to pay, they capture consumer surplus while the producer earns based on the gap between their costs and the market price. The only scenario where both producer and consumer surplus equal zero is when each consumer pays exactly their willingness to pay and the producer sells at exactly their minimum acceptable price. In this situation, the producer surplus and the consumer surplus would be zero, due to the fact that there is no difference between the willingness to pay and the actual price paid, or between the willingness to sell and the actual price received [4].

Critics may argue that using personalized prices is a benefit for companies because they are able to collect more profit from the consumer [5]. AI systems extract maximum willingness to pay from consumers, expanding producer margins and profits. Nevertheless, the supply and demand curve would boost the producer's margin, benefitting large companies. This destroys the balance between the supply and demand of goods drastically [9]. As companies gain more market power, many larger stakeholders control the market. This leaves the economy in a state of uncertainty, as if one of the larger stakeholders has a loss, the economy could result in a downfall [3]. Personalized pricing usage will experience growth alongside producer margins, as consumer welfare and the state of the economy declines[1]. (Fig. 2)

Source of Image: [12]



Ethical Implications

As widespread use of personalized pricing methods are adopted, many could argue that it causes price discrimination and privacy concerns. Some consumers may believe it causes injustice and is unfair to charge different prices differently depending on the availability of specific data. These are claims stating that personalized pricing is fair to everyone due to its adaptability to different situations and consumer needs. The high usage of data with minimal transparency leads to consumer privacy concerns. Due to the decreased privacy control, consumers are worried about their data being tracked by third-party sources. More concerns reveal that since consumer demographics are being used, there is no transparency on what they are being used for other than personalized pricing [10]. Some consumers have found targeted ways around this algorithm to set themselves up for lower prices. Using a VPN of a cheaper area is a common way to bypass the AI software, reducing the price due to the less common area or cheaper household income. Another way these algorithms are outsmarted is by placing a cheaper item in the online cart or delaying the time to purchase the product. The cheaper item lowers the starting price for other items, as the AI would believe that they are more likely to buy cheap products. On top of this, delaying the purchase time often results in the AI offering a discount to complete the purchase.

Conclusion

If personalized pricing continues to be widely adopted by businesses, our world could have a low consumer surplus, increased privacy breaching, and an overall economy almost entirely controlled by large companies. Artificial Intelligence may be extracting countless pennies from thousands of consumers every minute, using data without transparency, and fueling market shares of companies, all at the cost of consumers. Opponents may think that personalization benefits companies, while it leads to a downfall for both consumers and businesses. If personalized pricing is further developed in our society, we risk instability in privacy, pricing, and our overall economy.

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